



MARIST
UNIVERSITY®

BIOL 477L 114
BIOLOGY CAPPING
Spring 2026

Welcome to Biology Capping! This course is intended to help you synthesize material and skills from throughout your studies as a Marist Biology/Biomedical Sciences major and continue your evolution as a scientific thinker and communicator. We will discuss and critique recent findings from scientific literature and communicate them in both written and oral formats. Throughout the semester, we will discuss ethical implications and considerations relating to aspects of biology, methods for effectively communicating often-complex scientific concepts with the general public, and the role of research and science in the global community.

Instructor Information

Patrick Thieringer

Office Location: Donelly Hall 262

Email: Patrick.Thieringer@marist.edu

Office Hours

Monday 11:00AM-12:30PM | Thursday 10:00-11:30AM

Course Information

BIOL 477L BIOLOGY CAPPING

Credit Hours: 3

Meeting Location: DN268

Meeting Times: MR 200PM-315PM

Course Description

A capping course required for all Senior Biology, Biomedical Science and Biology Education majors. Topics vary. Offered every semester.

Pre-Requisite/Co-Requisite

A basic understanding of biology & chemistry is required, including BIOL130, 131 (C or better in both); CHEM 131, 132; 7 credits of 3/400- level BIOL coursework; and Senior status.

Course Learning Outcomes

COURSE GOALS AND LEARNING OBJECTIVES

In general, this course is designed to give you more challenging practice in scientific thinking. The course's major student learning objectives (SLOs) are organized into 4 overriding goals: I) You will feel like a scientist. II) You will demonstrate proficiency in reading and discussing primary literature. III) You will demonstrate proficiency in scientific writing. IV) You will be familiar with ethical research practices.

Your ability to meet these objectives will be assessed through classroom discussions, presentations, written assignments, and exams.

Goal I. You will feel like a scientist

SLO I-01: Students will be able to generate plausible and testable hypotheses given scientific data as well as distinguish the difference between hypotheses, theories, and laws.

SLO I-02: Students will be able to distinguish data from conclusions and critically assess and analyze data.

SLO I-03: Students will develop collaborative working skills and understand how scientific knowledge is advanced in a community environment.

SLO I-04: Students will develop the skill to think quantitatively and probabilistically, using inferential statistics

SLO I-05: Students will be able to apply scientific data to genuine questions and problems of medicine and biological research.

Goal II. Students will demonstrate proficiency in scientific writing

SLO II-01: Students will be able to summarize the format and general practices (figure legends, etc.) of research publications, including peer review.

SLO II-02: Students will be able to write technical papers or summaries with precision and conciseness consistent with scientific publications.

SLO II-03: Students will be able to cite and reference research articles in a manner consistent with scientific publication.

Goal III. Students will demonstrate proficiency at reading and discussing primary literature

SLO III-01: Student will demonstrate the ability to ask higher order questions based on primary literature.

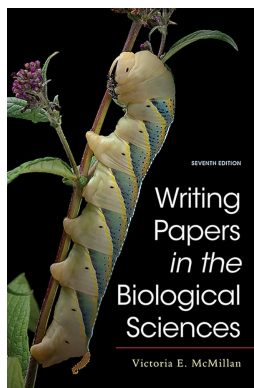
SLO III-02: Students, functioning as presenters, will be able to define precise learning objectives for their audiences.

SLO III-03: Student presenters will speak clearly, authoritatively, and engage their audiences in prepared and spontaneous presentations.

SLO III-04: Students will be able to keep discussions involving numerous participants focused and on task.

Goal IV. Students will be familiar with ethical research standards

Required Materials



Writing Papers in the Biological Sciences

Authors: Victor E. McMillan

Edition: 6th

Additional Information

The 5th and 7th editions are also fine.

Materials used in connection with this course may be subject to copyright protection.

Optional Resources

I will use Brightspace (Brightspace.marist.edu) as the course website to post announcements, quizzes, and share supplemental information (lecture notes, links, etc.). All announcements will be sent to your Marist email account, so be sure to check this email at least once per day.

Course Grade Calculation

Point Value Assignment
200 pt Final Exam
20 pt CITI RCR certification
20 pt Techniques presentation
50 pt Core paper
100 pt Attendance & Participation
100 pt Preparation for J. club
100 pt Lead J. club
10 pt Group topic selection
20 pt Topic introduction
15 pt Review paper - Annotated Bibliography
70 pt Review paper - Final
50 pt Experimental Design
30 pt Draft of topic project
20 pt Peer review preparation
20 pt Peer review summary
50 pt Final presentation
100 pt Final topic project
25 pt Peer evaluations
1,000 pt TOTAL

Final Course Grade Scale

Grading Scale	
Letter Grade	Percentage
A	1000-940
A-	939 - 900
B+	899 - 866
B	865 - 833
B-	832 - 800
C+	799 - 766
C	765 - 733
C-	732 - 700
D+	699 - 666
D	665 - 600
F	<599

Graded Assignments

FINAL EXAM (200 PTS)

There is a single exam in the course, a final exam that will assess your ability to interpret, analyze and draw conclusions from primary data. This exam will consist of short answer and essay questions. The best way to study is to actively participate in class discussions, especially journal clubs. You will need only your brain and a writing utensil to complete the exam. Make-up exams will be given only under exceptional circumstances (see Attendance Policy) and will require documentation.

CITI CERTIFICATION (20 PTS)

You will complete the Collaborative Institutional Training Initiative (CITI Program) training process for the Responsible Conduct of Research Basic (RCR) course designed for persons wishing to perform biomedical research. This training focuses on regulations and ethical considerations important in biological research and will inform in-class discussions. If you have completed this certification for another course, that documentation will suffice.

(<https://www.citiprogram.org/> to register for an account)

TECHNIQUE INTRODUCTION PRESENTATION (20 PTS)

There are a core set of techniques and tools that we'll likely see in many papers this semester; to ensure that everyone has an opportunity to understand what these techniques are and how they are used to answer biological questions, you'll work in small groups to present a brief (~10-15 minutes) introduction to an assigned technique. PDF of your presentation will be posted to our Brightspace site and be available as a reference throughout the semester if you need a refresher on a technique while reading a paper.

CORE PAPER (50 PTS)

The goals of the Core/LS Program are to develop values awareness, critical thinking, integrative learning, and writing proficiency. The goal of the core paper is for you to demonstrate all four of these competencies. You will read excerpts written by a prominent biologist where they discuss how they developed their passion for biology and how it changed them and their world-view. Your essay should be 3-5 pages and describe your own personal journey into biology. Additional details will be provided in class.

ATTENDANCE & PARTICIPATION (100 PTS)

Many of the learning objectives for this course center around discussing scientific literature, communicating as a scientist and understanding ethical considerations surrounding research and science. Therefore, the participation component of your grade will require not only your consistent attendance but also engaging in class discussions and journal clubs and asking and answering questions in class. If you have an unexcused absence, you will forfeit your opportunity for

participation points from that day. Participation points will not be adversely impacted by excused absences.

JOURNAL CLUB PREPARATION (100 PTS)

One of the major goals of this course is that students demonstrate proficiency in discussing the scientific literature. A successful discussion requires familiarity of content between all participants, so each student will be required to post questions/comments about each journal club article prior to class (due at 8pm the evening prior to class). These questions will be submitted via Brightspace. You are not required to comment on your own article (and cannot earn points for doing so). You can earn a maximum of 10 points per journal club, and a maximum of 100 points in this category, so you can either skip commenting on a couple papers or make up for partial credit earned for some journal clubs by commenting on all journal club papers.

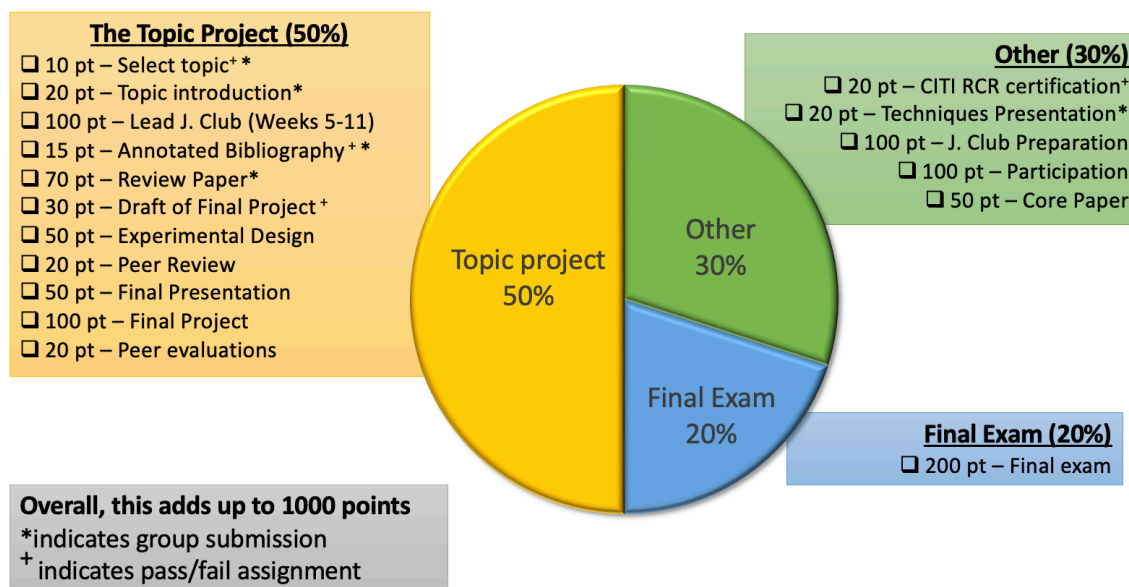
JOURNAL CLUB DISCUSSION LEADER (100PTS)

As the ability to integrate material from the primary literature into our knowledge of a topic is an important skill for working scientists, students will each present an article to their peers throughout the semester. These journal clubs will take place during the first 35-40 minutes of class and will occur in an informal round-table format (no slides other than figures from the paper). I will present the first paper so you have an idea what is expected from each presentation.

- For each journal club, ALL students are expected to read the paper prior to class. All articles must be submitted for approval by 2/2/26. Following approval of papers, I will post the PDFs on Brightspace under the appropriate dates.
- Journal club papers must serve as background material for your Biology Topic Project.

BIOLOGICAL TOPIC PROJECT ASSIGNMENTS (400 PTS)

The biological topic project is a group project and has been designed to create a more meaningful and relatable final project for biology students. Through a series of scaffolded assignments, you will produce resources for different audiences which are scientifically accurate yet convey information in approachable and meaningful ways. The full project will span the entire semester and include many components which are similar to the grant proposals produced by other students in biology capping each semester. Major assignments related to this project include writing a review paper, presentations, as well as creating original material to convey your topic to the public. Additional details will be presented in class.



Attendance Policy

If you're sick, please don't come to class and share your illness with everyone else. I will be as flexible as possible around in-class attendance, but communication is

key if you will miss/think you may miss a class. Attendance is expected if you are healthy, but please respect your peers and everyone in the campus community by staying home (or wearing a mask in class) if you're sick.

Lecture: You are encouraged to attend all class meetings. All content is available on the course website and I'm happy to address questions during office hours if you cannot make it to class. Additionally, assessments can be delivered via Brightspace, so you will be able to turn in work, take quizzes, etc. from outside of the classroom and not fall behind if you are out of class for a few days.

Academic Support

Marist University provides a range of resources to help you get the most from your educational experience. These resources include but are not limited to tutoring, writing assistance, advising, proofreading assistance, and counseling. A full list of these resources can be found at this link. <https://www.marist.edu/academic-resources>

Technical Support and Support Links

Classrooms and Labs support services are managed through the IT Help Desk and Media Center. Computer related problems should be reported by calling the Marist University IT Help Desk at (845) 575-4357 (xHELP) or email: helpdesk@marist.edu. Problems with overhead projectors or other media related equipment should be reported to the Media Center Help Desk at (845) 575-3635 (x3635) or email: media@marist.edu.

To access Brightspace, visit: <https://brightspace.marist.edu/d2l/home> and click on "Marist SSO" to login directly with your Marist credentials. For helpful Brightspace tool guides and tutorials, visit: <https://my.de.marist.edu/teaching-technologies>

Statement on Academic Honesty

Marist University is a learning community dedicated to helping students develop the intellect, characters, and skills required for enlightened, ethical, and

productive lives in the global community of the 21st century. Students are expected to pursue excellence in their education while being honest about their work and fair to other members of the learning community. All work presented to instructors for evaluation must reflect their own ideas and effort and must properly acknowledge any contributions of others. Students should expect this honesty and fairness in others as well. As members of the Marist learning community, all students should adhere to the principles of academic integrity as set forth in the Marist Academic Integrity Policy.

<https://www.marist.edu/student-life/community/community-standards/academic-integrity-policy>

Detection Software Statement

Use of software that detects plagiarism, including AI generated content, is mandated in this course. Copyleaks is a service used by Marist University faculty to compare a student's work with its very large database of sources, student papers from other institutions, and the like, to check for originality. Work submitted to Copyleaks will be used only for the purposes of assessing originality, and will not be shared beyond Copyleaks or used for any other purpose. Students must submit all assignments to Copyleaks through regular Brightspace submission process. If your instructor enables Copyleaks, there will be an End User License Agreement (EULA) that you must accept. It will appear to you on the assignment page before the area where you submit the assignment. You can view and accept the EULA by clicking on the link and then clicking "Finish" to accept the terms of use. Students who wish to remove their personal identifying information (name, student identification number, etc.) from the submitted file may do so but must notify their professor ahead of the submission. Work submitted through Brightspace in this course will not be reviewed by the Professor or maintained by the University unless and until the Copyleaks process is completed.

For more information about Copyleaks visit: <https://my.de.marist.edu/copyleaks>

Accommodations and Accessibility

From the Office of Accommodations and Accessibility

Students with disabilities who believe they may need accommodations in this class are encouraged to contact the Office of Accommodations and Accessibility at (845) 575-3274, Donnelly Hall 226 or via email at accommodations@marist.edu as soon as possible to better ensure that such accommodations are implemented in a timely manner. Marist's guidelines for instructors to comply with the Americans with Disability Act (ADA) are located here: <https://www.marist.edu/student-life/community/accommodations-accessibility/guidelines-instructors>

Your Privacy Rights - Family Education Rights and Privacy Act

The Family Educational Rights and Privacy Act (FERPA) (20 U.S.C. § 1232g; 34 CFR Part 99) is a Federal law that protects the privacy of student education records. Marist University collects and retains data and information about students for designated periods of time for the expressed purpose of facilitating the student's educational development. The University recognizes the privacy rights of individuals in exerting control over what information about themselves may be disclosed and, at the same time, attempts to balance that right with the institution's need for information relevant to the fulfillment of its educational missions. As employees of the University, faculty, administration and staff all have some access to student record information. Please, visit <http://www.marist.edu/registrar/ferpa.html> for more information on how MaristUniversity protects your privacy rights.

Copyright Information

Materials in this course may be subject to copyright protection.

Title IX Statement

Marist University is committed to providing a safe learning environment for all students. If you or someone you know has experienced sexual harassment, including sexual assault, dating or domestic violence, or stalking, support is available. Please contact the Title IX Office at titleix@marist.edu or (845) 575 - 3799 or visit www.marist.edu/title-ix to file a report. Please be aware that faculty and staff are required to disclose incidents of sexual harassment or other potential violations of the Marist University Discrimination, Harassment, and Sexual Misconduct Policy to the Title IX Office. To speak to a confidential resource who does not have this reporting responsibility, contact Counseling Services at (845) 575 - 3314, Health Services at (845) 575 - 3270, or Department of Spiritual Life & Service at (845) 575 - 3000 (x2275). For more information about reporting options and resources at Marist University, please visit <https://www.marist.edu/title-ix>

Community and Belonging Statement

The university's academic mission is immeasurably enriched by students with diverse experiences. Our finest efforts as intellectual beings heavily rely on the exchange of ideas. Interactions in our classrooms among persons and groups with diverse backgrounds, ideologies, and experiences facilitate these efforts by allowing us all to be more reflective about the varied historical and social contexts in which we work and learn. For faculty and students to continue being leaders inside and beyond academia, we must ensure that we consider the diversity of all who comprise our communities and foster a climate in which those diverse influences are respected and valued. In this course, we will challenge each other's thinking while working collaboratively to ensure that the classroom is a space of safety and bravery. Our classroom offers an environment where individuals of varying opinions, experiences, and backgrounds are able to be free to learn without fear of being silenced. Evidence of these efforts will manifest in readings, lectures/class discussion, seminars, and group projects. Aspects of diversity include, but are not limited to, race, ethnicity, color, nationality, sex, gender, gender identity, gender expression, class, sexual orientation, religion, age, ability, and veteran status. Students who would like to be identified in a manner other

than what is indicated on the course roster can contact me privately via phone, email, web conference or face-to-face meeting to indicate name, pronoun and any other preferences they may have.

Additional Course Policies

LATE POLICY

Most assignments (not exams) have a penalty-free 48-hour grace period from the submission deadline. Any assignments which do not have this grace period will be clearly indicated on the class website.

- Questions about assignments will not be answered during the grace period, or after 5pm on the original due date of the assignment.
- Assignments will NOT be accepted 7 days after the due date and you will receive a zero for that assignment

Following the 48-hour grace period:

- Assignments submitted up to 24 hours after the end of the grace period will incur a 10% penalty.
- An additional 10% penalty will be applied for each additional 24hr period.

EXCUSED ABSENCE POLICY

The following constitute the only valid excuses for missing class, quizzes, exams, assignments or in-class activities:

1. Death in the immediate family. Present a note from the Center for Advising & Academic Services.
2. Officially-approved College activities such as intercollegiate athletic events or theatrical productions. Present a note from the appropriate College official stating that your participation in this activity prevented

you from being in class at the exact date and time it was held.

3. Illness. If you're sick or think you might be getting sick, STAY HOME. As noted above, all assessments and assignments can be completed online, so you are expected to stay on track with quiz dates, assignment due dates and other components throughout the semester whether you are in the classroom or remote.
1. Debilitating illness. If you are too ill to complete quizzes/assignments/etc. on time, I will require documentation stating that the illness prevented you from participating in class remotely at the exact date and time that the quiz/assignment and/or laboratory was held. If you seek medical attention at Marist Health Services, you must provide health service with documented authorization allowing me to verify your illness or injury. A note that simply states that you were under the care of a physician, or had an appointment with a doctor, is not sufficient.

If an absence is excusable and of sudden or unpredictable nature, I still require notification as soon as possible (at the latest: on the day you return to class; preferably prior to that date, via email) in order to reschedule missed assessments. Please note that it is your responsibility to schedule a make-up when you return to class. If a make-up is not rescheduled within one week of your return to class, you will have lost your opportunity to do so.

Course Calendar

Date	Topics & Journal Club Schedule		Assignment/Readings
1/22	Course introduction and overview		
1/26	Group Topic Assignment Introduction		
1/29	Library and article discussion with Elizabeth Clarke		CITI RCR certification (Wed 1/28)
2/2	Core paper discussion		Topic approval (Thu 2/2)
2/5	Scientific writing and communication	Reading and analyzing scientific literature	
2/9	Review Papers: In-class exercise		J. Club paper approval (2/11)
2/12	Techniques presentations		Technique presentation (2/12)
2/16	Demo Journal club	What can you do with a biology degree?	Demo J. club reading (2/16)
2/19	Project topic introduction presentations		Topic Introduction ppt (2/19)
2/23	Group 1: Journal club 1	Ethics in Science	
2/26	Group 1: Journal club 2	Ethics in science cont.	
3/2	Group 1: Journal club 3	Analyzing articles - Example	Annotated bibliography (3/8)
3/5	Group 2: Journal club 1	Ethics in Science – Applied Example	Ethics Applied Example Reading (3/5)
3/9	Group 2: Journal club 2	Group 2: Journal club 3	Core paper (3/13)
3/12	Group 3: Journal club 1	Fact checking in science	
SPRING BREAK – No class (3/13-3/22)			
3/23	Communicating Science: target audience	Designing scientific summaries for the public	
3/26	Group 3: Journal Club 2	Group 3: Journal Club 3	Review Article (3/29)
3/30	Group 4: Journal club 1	Communicating Science cont.	
4/2	Group 4: Journal club 2	Group 4: Journal club 3	
4/6	Group 5: Journal club 1	Ethics case study	Ethics Case Study reading (4/6)
4/9	Group 5: Journal club 2	Group 5: Journal club 3	
4/13	Experimental design assignment		Draft of project (4/16)
4/16	Science in the media and public perception		
4/20	What is Peer Review?		
4/23	In-class peer review day		Peer review prep (4/23)
4/27	Bias in Science communication		Peer review summary (4/26)
4/30	End of class wrap-up and assessments		Experimental design (5/3)
5/4	Topic Presentation		
5/7	Topic Presentation		Peer Evaluations & Final Project (5/7)
5/14	Final Exam: Thursday May 14 th at 1:00 PM		

(Schedule is subject to change)

General Assistance & Study Suggestions

GENERAL ASSISTANCE & STUDY SUGGESTIONS

This course will require a significant amount of time spent preparing for class meetings and discussions outside of class, as well as planning, organizing, writing and editing written assignments. In order to be successful, you must read all journal articles and other assigned readings before coming to class and be ready to contribute to discussions during class.

The major written assignments for the course (grant proposal, core paper) will be graded on both scientific merit/organization as well as proper grammar and logic, so be sure to edit your work and seek assistance from outside sources (i.e. the Marist Writing Center) if you have any doubts about the organizational/grammatical aspects of your work.

Active participation in class discussions, including asking questions, commenting on material being covered and responding to questions asked by myself and your peers, will improve your understanding of material and will be reflected in your final grade.

Lastly, if you have questions regarding any of the reading, including data in your journal club paper, stop by during office hours or contact me via email; I'm happy to go over material outside of class to help improve your understanding.

EXTRA CREDIT

While I may offer extra credit questions on quizzes or other assignments, I will not give you a personal extra credit assignment. Please do not ask me for an extra credit assignment. If you are concerned about your grade, please contact me with questions that directly relate to your understanding of the material. At the end of the semester, you will receive the grade that you've earned based on the point breakdown explained in this syllabus.